



CONTROLLED ENTERPRISE DOCUMENT

TBT-01.18 | Confined Space Entry Rebuilt

MH Construction, INC Enterprise Management System

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Use only after required functional, technical, field-usability, and approval gates are complete.

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MH Construction, INC

TOOLBOX TALK: TBT-01.18

TOPIC: Confined Space Entry — Atmospheric Hazards & Pasco Seasonal Patterns

TRACK: Track 3 MEP & Interior | Track 1 Civil & Earthwork

OSHA: 29 CFR 1926.1207 | WISHA: WAC 296-809

CSI Division: 12 / 22 / 23 / 31 / 33

Prerequisite: TBT-01.16 Wind Safety & Crane Limits

Cross-Reference: MISH-F-32.1 Confined Space Permit

⚠ CRITICAL: Confined space fatalities occur in clusters. When one worker collapses from an atmospheric hazard, untrained rescuers enter and also die. The attendant NEVER enters. Call 911 first. Non-entry retrieval is the ONLY authorized rescue.

1. DIRECTIVE (Policy Standard)

No MH Construction, INC employee shall enter a permit-required confined space without a signed MISH-F-32.1 Confined Space Permit, a calibrated 4-gas monitor actively sampling the space, and a designated Attendant stationed outside the space for the entire duration of the entry. Atmospheric testing must occur BEFORE the lid or hatch is fully removed to prevent flash ignition or exposure to the opening crew.

2. GENERAL ORDER (Execution Standard)

The Tri-Cities region presents unique seasonal atmospheric hazards that turn standard confined spaces into IDLH (Immediately Dangerous to Life or Health) environments. The Foreman and Attendant must adjust their monitoring strategy based on the current season:

Season	Atmospheric Hazard	Pre-Entry Action
Spring (Mar–May)	Irrigation season groundwater intrusion — H ₂ S from disturbed sediment	Lower monitor to bottom, wait 3 min. Pump standing water before testing.
Summer (Jun–Aug)	100°F+ heat accelerates O ₂ depletion in sealed spaces (rust/oxidation)	Monitor every 15 min. Limit entry to 20 min. Forced ventilation mandatory.
Fall (Sep–Nov)	Wildfire/agricultural smoke introduces CEO that concentrates in low spaces	Check ambient CEO before opening. If >9 ppm, postpone entry.
Winter (Dec–Feb)	Methane migration from frozen ground in utility corridors	Test LEL before and continuously during entry. No open-flame tools inside.

A 4-gas monitor tests for Oxygen (O₂), Hydrogen Sulfide (H₂S), Carbon Monoxide (CO), and Lower Explosive Limit (LEL). The monitor must be bump-tested daily and lowered slowly into the space to test all levels (top, middle, bottom) because gases stratify based on their weight.

3. KEY HAZARDS & MITIGATION (The "What If")

SCENARIO 1: Manhole H2S Stratification

Work Type / Season: Div 33 Utilities | Spring (Irrigation Season)

The Hazard: A crew opens a sewer manhole and drops the monitor just inside the rim. It reads clear. The entrant descends 8 feet to the bottom and collapses immediately. Hydrogen Sulfide (H2S) is heavier than air and has pooled entirely at the bottom of the manhole.

The Mitigation: A surface reading of zero does not mean the bottom is safe. The monitor must be lowered on a rope, pausing for 30 seconds at every 4 feet of depth to allow the sensors to process. Never trust your nose — H2S smells like rotten eggs initially, but quickly deadens your sense of smell (olfactory fatigue).

TBT-01.19 Bridge: *TBT-01.19 Scenario 1 covers the fall hazard on the icy rungs of this same manhole. If the entrant falls and is incapacitated in the H2S layer, the retrieval system is the only way to recover them without a second fatality.*

SCENARIO 2: Sealed Mechanical Room O2 Depletion

Work Type / Season: Div 23 HVAC / Div 22 Plumbing | Summer

The Hazard: A mechanical room in a basement has been sealed for a month. The steel pipes inside have been rusting. The rusting process (oxidation) consumes oxygen from the sealed air. A plumber enters to check a valve and begins suffocating because the O2 level has dropped below the 19.5% minimum threshold.

The Mitigation: Any sealed room closed for 48+ hours is a confined space until tested. Do not enter immediately upon opening the door. Set up a blower and force-ventilate the room for 5 minutes, then test the air before crossing the threshold.

TBT-01.19 Bridge: *TBT-01.19 Scenario 2 covers harness entanglement in this pipe-congested room. Heat exhaustion combined with low O2 accelerates incapacitation. The retrieval system must be capable of lifting a fully limp worker through the congestion.*

SCENARIO 3: Plumbing Pit Methane + Torpedo Heater

Work Type / Season: Div 22 Plumbing / Div 31 Earthwork | Winter

The Hazard: A crew is working in a deep plumbing pit in December. It is 20°F outside, so they lower a propane torpedo heater into the pit to stay warm. Methane gas migrating from the frozen soil enters the pit and is ignited by the heater, causing a flash fire.

The Mitigation: Torpedo heaters and internal combustion engines are strictly prohibited inside confined spaces. They consume oxygen, produce carbon monoxide, and provide an ignition source for LEL hazards. Heat the space by ducting warm air in from a heater located at least 25 feet away on the surface.

SCENARIO 4: Utility Vault Wildfire CEO Concentration

Work Type / Season: Div 33 Utilities | Fall (Wildfire Season)

The Hazard: During a severe AQI smoke event, ambient Carbon Monoxide (CO) settles into a low-lying utility vault. The crew tests the air and finds CEO at 25 ppm. They put on N95 masks and prepare to enter.

The Mitigation: An N95 mask filters particulates (smoke ash) but provides ZERO protection against gases like Carbon Monoxide. If CEO exceeds 9 ppm, the space must be force-ventilated until clear. If it cannot be cleared, entry requires a Supplied Air Respirator (SAR) — not an N95.

TBT-01.19 Bridge: *TBT-01.19 Scenario 3 covers retrieval tripod failure on soft asphalt. If the entrant is overcome by CEO and the tripod collapses during rescue, the situation escalates to a multi-fatality event.*

SCENARIO 5: PEMB Foundation Pit — Dual Hazard

Work Type / Season: Div 13 Special Construction / Div 31 Earthwork | All Seasons

The Hazard: A crew digs a 10-foot deep, narrow foundation pit for a PEMB structural column. A gas-powered generator is running 10 feet from the edge. Exhaust fumes sink into the pit. The pit walls are vertical and unshored.

The Mitigation: Pits deeper than 4 feet with restricted access are BOTH confined spaces AND excavations. You must follow both protocols. Shoring or benching is required for the walls, AND atmospheric testing is required for the air. Keep all combustion engines at least 25 feet downwind of the opening.

TBT-01.19 Bridge: *TBT-01.19 Scenario 4 covers the wind updraft hazard when descending into this open pit. The SRL must be clipped in before the first step onto the ladder.*

SCENARIO 6: Crawl Space Fumigant Residue

Work Type / Season: Div 03 Concrete / Div 06 Wood | All Seasons (Agricultural Sites)

The Hazard: A crew enters a crawl space beneath a former food processing facility. The space was fumigated years ago, but chemical residue remains in the dry soil. Crawling disturbs the dust, releasing the chemical into the breathing zone.

The Mitigation: Standard 4-gas monitors do not detect complex agricultural chemicals or fumigants. If the site has an agricultural or processing history, a PID (Photoionization Detector) may be required. Treat unknown agricultural crawl spaces as contaminated until cleared by the Safety Officer.

TBT-01.19 Bridge: *TBT-01.19 Scenario 5 covers horizontal drag rescue in this exact crawl space. If the entrant is overcome by fumigants at 35 feet, the retrieval line must be long enough to pull them out.*

4. FIELD INSTRUCTION (Immediate Action)

Before any confined space entry today, execute this 5-point atmospheric check:

- (1) Pull MISH-F-32.1 Confined Space Permit — no permit, no entry.
- (2) Lower the 4-gas monitor to the bottom of the space and wait 3 full minutes before reading.
- (3) Confirm the Attendant is stationed at the opening with the rescue plan and 911 ready.
- (4) Apply the Pasco seasonal atmospheric hazard mitigation for today's conditions.
- (5) Ensure NO combustion engines are operating within 25 feet of the opening.

If the monitor alarms at any point during the entry — EVACUATE IMMEDIATELY. Do not stay to finish the task. Do not try to find the source. Get out. If an entrant reports dizziness, headache, nausea, or visual disturbance — EVACUATE IMMEDIATELY.

5. ATTENDANCE LOG

Foreman: Scan the QR code below to log attendance digitally via MISH-F-10.1 on the MH Dashboard, or use the physical sign-in sheet in the site binder as the fail-safe. File the completed MISH-F-32.1 Confined Space Entry Permit together with this attendance log.

[INSERT QR CODE HERE — LINK TO MISH-F-10.1 ON MH DASHBOARD]

SIGN-IN SHEET (Hardcopy Fail-Safe)

Project: _____ Date: _____ Foreman: _____

Employee Name (Print)	Role (Entrant / Attendant / Supervisor)	Monitor Bump Tested (Y/N)	Signature